Ofqual QAN	**600/6407/9 / 603/0080/2**
Quartz ID	**1047 / 1125**
Qualification	**Level 3 Diploma in Computing / Level 3 International Foundation Diploma for Higher Education Studies**

Assessment	**Introduction to Computer Science 2.0**
Assessment Type	**Examination**
Date	**Wednesday 4th June 2025**
Time	**9.00am – 12.00pm**

Examination Paper

Answer ALL questions.

Clearly cross out any surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Section A – Multiple Choice Question
Answer these in your answer booklet – do NOT write on this Question Paper.
Write ONE (1) correct answer from A, B, C, or D for each question in your answer booklet.
Each question is worth 1 mark.

Question 1

What is the key difference between primary and secondary storage?

A Primary storage is non-volatile, while secondary storage is volatile

B Secondary storage is faster than primary storage

C Primary storage is used for temporary data storage, while secondary storage is used for long-term data storage

D Secondary storage cannot store files larger than a gigabyte

1 mark

Question 2

The hexadecimal code FF, can be written in binary as which code?

A 1515

B 1111

C 0000 0000

D 11111111

1 mark

Question 3

Which protocol is responsible for ensuring that data is reliably and accurately transmitted across the Internet?

A HTTP

B FTP

C SMTP

D TCP/IP

1 mark

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Marks

Question 4

Which component of the CPU is responsible for fetching, decoding and executing instructions?

A Arithmetic and Logic Unit (ALU)

B Memory Data Register (MDR)

C Accumulator

D Control Unit (CU)

1 mark

Question 5

What is a key challenge in implementing global cybercrime laws?

A The rapid development of new cyber threats

B The lack of interest from governments

C The unwillingness of individuals to follow laws

D The absence of international trade agreements

1 mark

Question 6

Why might an organisation choose a Virtual Private Network (VPN) over a public network?

A To increase the speed of the network

B To replace firewalls and antivirus software

C To connect without using internet protocols

D To secure data transmission and protect privacy

1 mark

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Marks

Question 7

Which of these statements is **NOT** correct?

A A mouse is an input device

B A speaker is an input device

C A keyboard is an input device

D A printer is an output device

1 mark

Question 8

Which type of storage is most suitable for backing up large amounts of data in a business environment?

A Solid-state drive (SSD)

B Magnetic tape

C Optical disc (CD/DVD)

D USB memory stick

1 mark

Question 9

Why is cache memory important for a CPU's performance?

A It provides long-term storage for programs and files

B It increases the processor's clock speed

C It stores frequently accessed data closer to the CPU for faster retrieval

D It controls the execution of arithmetic and logical operations

1 mark

Question 10

Where is HTML primarily used in web development?

A In storing data on web servers

B In styling the appearance of webpages

C In defining the structure and content of webpages

D In managing data transmission between computers

1 mark

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Marks

Question 11

Why is an embedded system different from a general-purpose computer?

A Embedded systems can only connect to the internet

B General-purpose computers cannot store data permanently

C Embedded systems are designed for a specific function within a device

D General-purpose computers do not require an operating system

1 mark

Question 12

A gaming company is developing a new high-performance computer for competitive gaming. They need to ensure the best processing speed and efficiency for real-time gameplay.

Which type of processor architecture would be most suitable?

A Single-core processor

B Dual-core processor

C Multi-core processor

D RISC processor

1 mark

Question 13

Where is system software typically stored on a computer?

A In a removable USB drive

B In the cloud

C In the ROM or hard drive

D In the CPU’s arithmetic and logic unit

1 mark

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Marks

Question 14

What is the key difference between hardware and software?

A Hardware is used for storage, while software is used for processing

B Hardware is a physical component, while software consists of instructions that run on a computer

C Software is always more expensive than hardware

D Hardware cannot function without an internet connection

1 mark

Question 15

Which language is the code **<h1> Welcome </h1>** an example of?

A HTML

B CSS

C Java Script

D Python

1 mark

Question 16

Which of the following is an example of a tactile output device?

A Printer

B Speaker

C Braille display

D Projector

1 mark

Question 17

Where is a Metropolitan Area Network (MAN) most commonly used?

A In a single building such as an office or school

B To link networks across a city or university campus

C For personal device connections, such as mobile phones and tablets

D To connect different countries via satellites

1 mark

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Marks

Question 18

How is the NOT gate different from other logic gates?

A It requires multiple inputs to function

B It always produces a high output

C It has only one input and inverts the signal

D It operates without using binary logic

1 mark

Question 19

Which of the following best describes utility software?

A A type of software that helps to maintain, secure, and optimise a computer system

B A collection of programs that perform mathematical calculations

C Software that allows users to write documents and spreadsheets

D Software that only runs on mobile devices

1 mark

Question 20

Why is the Digital Divide considered an ethical issue in computing?

A It leads to economic inequality between those with and without technology access

B It increases the risk of cybercrime

C It prevents people from using social media effectively

D It only affects developing countries

1 mark

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Marks

Question 21

A graphic designer is creating a company logo that will be printed on business cards, billboards and used on a website. Which type of image format should they use, and why?

A Bitmap (JPEG) – High quality and supports millions of colours

B Vector (SVG) – Scalable without losing quality suitable for multiple sizes

C GIF – Small file size and supports animation

D PNG – Supports transparency and high-quality images

1 mark

Question 22

Why are SSDs generally considered more reliable than HDDs?

A They do not require an operating system to function

B They use less power and do not generate heat

C They are cheaper to produce

D They have no moving parts, making them more durable

1 mark

Question 23

Where is the arithmetic and logic unit (ALU) located within a computer system?

A In the motherboard’s chipset

B Inside the hard drive

C Within the RAM module

D As part of the CPU

1 mark

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Marks

Question 24

The owner of a small business is looking for software that will allow employees to write reports, create spreadsheets and prepare presentations. However, they also need to ensure that different employees can collaborate on the same documents online.

Which type of software would be the best choice for their needs?

A Standalone word processing software

B Operating system software

C Antivirus software

D Integrated office suite (e.g., Microsoft 365, Google Workspace)

1 mark

Question 25

Who should an individual go to if they believed their personal data was being misused under the UK's Data Protection Act?

A The UK Parliament

B The Information Commissioner's Office

C The United Nations Cybersecurity Division

D The Police

1 mark

Question 26

Which of the following would **NOT** affect the quality of an outdoor digital sound recording?

A The level of background noise

B The bit depth

C The sample rate

D The amount of file storage needed

1 mark

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Marks

Question 27

A security system for a warehouse uses a logic circuit to trigger an alarm. The system is designed so that the alarm will only activate if both a motion sensor and a door sensor detect an intruder.

Which type of logic gate is most suitable for this security system, and why?

A NOR gate – Ensures the alarm activates when neither sensor detect

B OR gate – Ensures the alarm activates if either sensor detects an intruder

C NAND gate – Ensures the alarm is off unless both sensors detect an intruder

D AND gate – Ensures the alarm only activates when both sensors detect an intruder

1 mark

Question 28

What is a key disadvantage of a mesh network topology?

A If one device fails, the entire network stops working

B It requires an internet connection to function

C It is expensive and complex to install

D It does not allow direct communication between devices

1 mark

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Marks

Question 29

Sarah is setting up a new business and needs to decide on a network type. She wants a secure system where all employees' data is stored in one place and access can be controlled by an administrator. However, she also wants employees to be able to share files efficiently.

Which type of network would best suit Sarah's business?

A LAN

B P2P

C Client-Sever

D WAN

1 mark

Question 30

Which of the following is an example of a cultural issue in computing?

A Different countries have varying privacy laws regarding online data storage

B Hacking is a growing crime

C Using copyrighted software without a license

D The environmental impact of manufacturing computer hardware

1 mark

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Marks

SECTION B
ANSWER ALL QUESTIONS

Question 31

a) State the **FOUR (4)** main functions of a computer system. **1**

b) Explain the difference between system software **and** application software, providing **ONE (1)** example of each. **3**

c) State **THREE (3)** types of operating system interface **and** explain how each **ONE (1)** is used. **3**

d) Discuss the advantages **and** disadvantages of proprietary (closed-source) software compared to open-source software. **3**

Total 10 Marks

Question 32

a) State the smallest unit of data storage in a computer system. **1**

b) Explain the difference between how magnetic, optical **and** solid-state storage store data, giving **ONE (1)** advantage of each. **3**

c) Describe **THREE (3)** different types of input device **and** provide **ONE (1)** example of where each would be used in library computer system. **3**

d) Describe the factors that should be considered when selecting an output device for a computer system. **3**

Total 10 Marks

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Marks

Question 33

a) State the primary reason computers use binary instead of decimal. **1**

b) Convert the decimal number 45 into binary **and** hexadecimal. **3**

c) Explain how two's complement is used to represent the number -10 in an 8-bit system. **3**

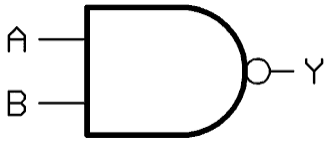
d) Describe the key differences between lossy **and** lossless compression for sound **and** provide an example of when each would be appropriate. **3**

Total 10 Marks

Question 34

a) A warehouse is protected at night by an alarm system (Y). A and B are **TWO (2)** external light sensors that are used to turn off the alarm when the sun rises. A sensor gives a signal TRUE when it is in sunlight. **4**

Draw **and** complete a truth table for the following logic gate for the alarm system (Y) **and** describe the conditions when the alarm system is active (TRUE).



b) State the role of the system bus in a computer. **1**

c) Explain the **THREE (3)** stages of the fetch-decode-execute cycle. **3**

d) Describe **ONE (1)** difference between RAM **and** ROM. **2**

Total 10 Marks

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Marks

Question 35

a) State the main function of a repeater in a computer network. **1**

b) Describe **THREE (3)** types of network transmission media. **3**

c) Explain **THREE (3)** differences between the TCP/IP **and** OSI models in network communication. **3**

d) Describe the purpose of a MAC address **and** give an example of how it is represented. **3**

Total 10 Marks

Question 36

a) Define the term ‘malware’? **1**

b) Other than **malware**, state **THREE (3)** types of cyber threat that can affect a computer network. **6**

Describe **ONE (1)** way to prevent each threat.

c) Explain the difference between an intranet **and** an extranet. **3**

Provide **ONE (1)** example of where each might be used.

Total 10 Marks

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Marks

Question 37

a) The Computer Misuse Act (1990) outlines several offences related to cybercrime. **4**

Identify **and** explain **TWO (2)** specific offences covered under this Act **and** provide an example of each.

b) Many companies rely on automation in the workplace, but this raises ethical and environmental concerns. **6**

Discuss both the ethical **and** environmental impacts of automation in business operations.

Total 10 Marks

End of paper

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